Searching for the secrets of the early Universe

MPhys Public Summary

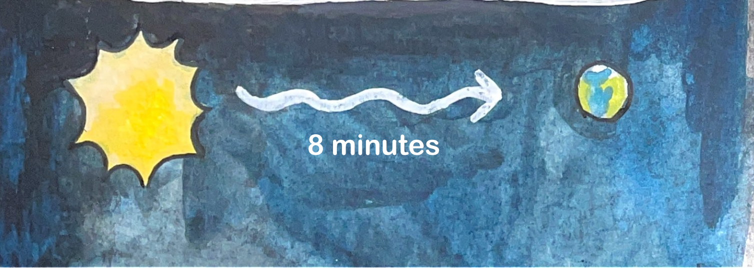
Clara Pollock 2024

This is our galaxy - The Milky Way.
It is full of stars, planets, and gas, and it is held together by gravity.

There are billions of galaxies in the Universe - everywhere we look in the sky!

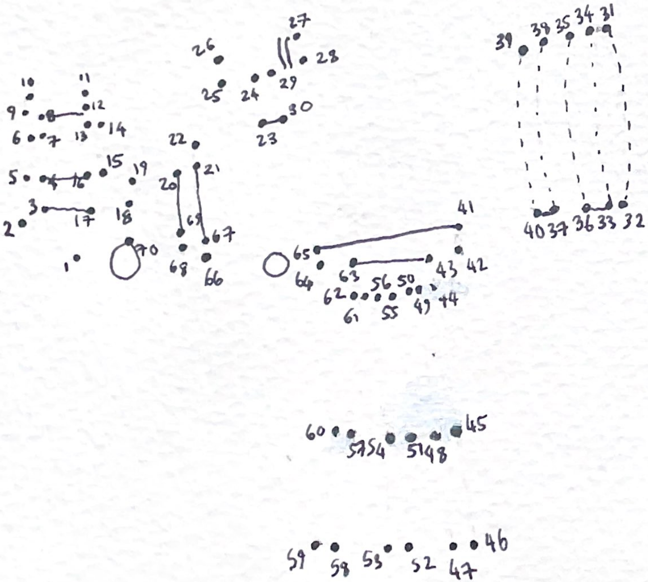
There are lots of galaxies now, but something that interests a lot of scientists are the FIRST stars and galaxies in the universe, and how they are different from our Milky Way.

The Universe is expanding, so these galaxies are very far away. Light travels fast, but it takes 8 minutes to get to Earth from the Sun, and billions of years from these distant galaxies. So we can see galaxies that were born not long after the start of the universe!

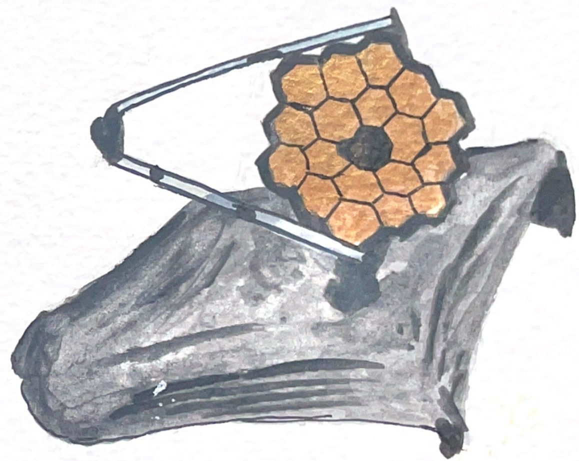


If you have ever looked through a pair of binoculars, you will know that you can see things that are far away much bigger and clearer with them.

The same is true with telescopes!
They let us see stars and galaxies
that are far away!
Can you complete the dot-to-dot to
reveal the telescope?

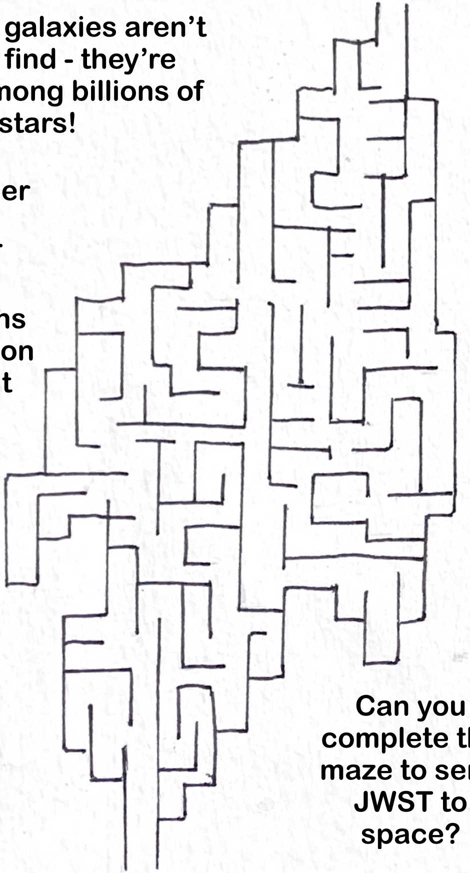


The telescope that is used to look for early
galaxies is the James Webb Space Telescope
(JWST).
JWST is very special because it can see things
no other telescope can.
JWST can see in infrared, which helps it see
through dust clouds to find the early galaxies!



Our early galaxies aren't easy to find - they're hidden among billions of stars!

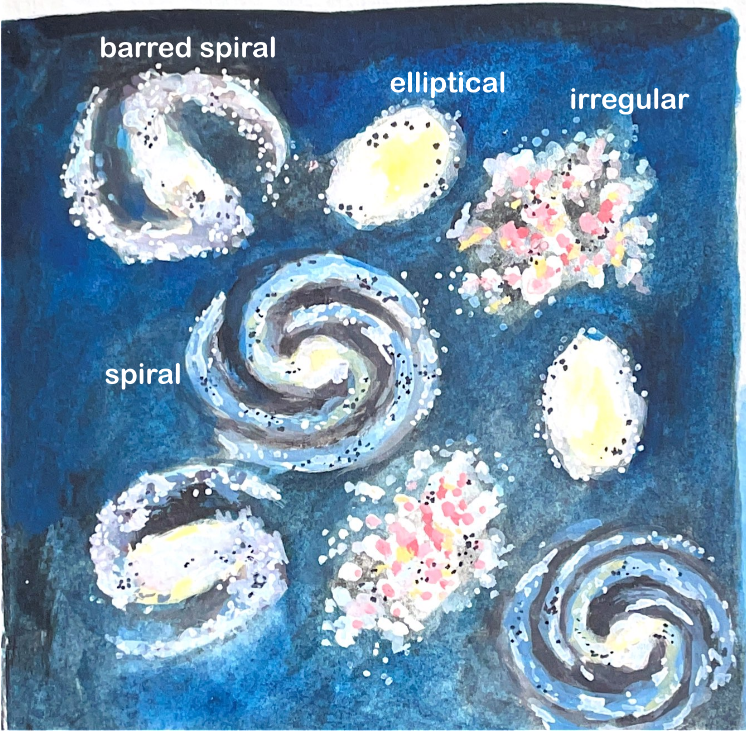
It was super important that JWST had no distractions from light on Earth, so it had to be sent to space!



Can you complete the maze to send JWST to space?

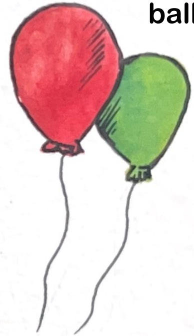
Why do we want to find these early galaxies? Well, they might hold clues about how our own galaxy formed, and how it has changed over billions of years.

What about shapes? In the early Universe galaxies were still forming, growing, and changing. How did they form into the beautiful shapes we see today?
Can you pair up the galaxy shapes?

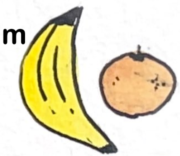


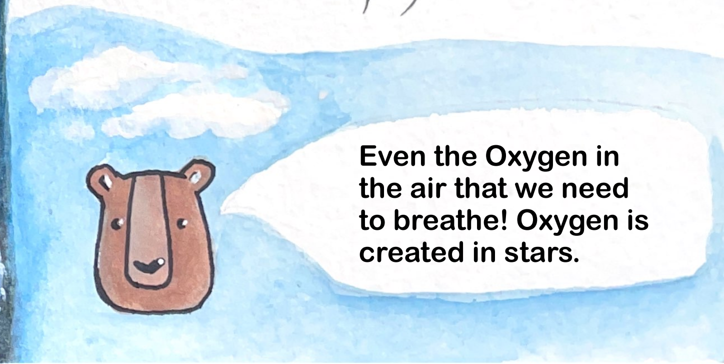
Another important thing is the chemical elements. Everything we see is made up of different elements.

Like the Helium that makes balloons float!



Or Potassium in fruits and veg!





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For my niece and nephew
Freya and James